

MARSHLLING

MARSHALLING OF MAIL / EXPRESS / PASSENGER TRAINS

Marshalling refers to arranging or attaching coaches in a predetermined, logical order to:

- ✓ Ensure train safety
- ✓ Improve operational convenience
- ✓ Provide maximum passenger ease
- ✓ Enable efficient handling of parcels, mails, and luggage

DETAILED NOTES ON ATTACHMENT OF DEAD LOCOMOTIVES

◇ Definition

A **Dead Locomotive** is one which is **not under power** (traction motors and prime mover not working) and **not providing tractive or braking effort** but needs to be hauled by another locomotive for operational, maintenance, or transfer reasons.

◇ Conditions for Attachment of Dead Locomotive

(i) Fit-to-Run Certificate

Before attaching a dead locomotive to any train, a '**Fit-to-Run**' certificate must be issued by one of the following:

- Section Engineer (Loco)
- Loco Inspector
- Power Controller (in coordination with shed)

This certificate ensures the dead locomotive is **mechanically and electrically safe** for movement.

(ii) Speed Compatibility

The **maximum permissible speed (MPS)** of the train must not exceed the **MPS** of the dead locomotive.

Example: If a dead loco is restricted to 75 km/h, a 110 km/h Mail/Express must reduce its speed to 75 km/h when hauling it.

(iii) Brake Synchronization

- The **brakes of the dead locomotive must function in synchronization** with the brakes of the working locomotive/train.
- Air brake continuity through **Brake Pipe (BP)** and **Feed Pipe (FP)** must be ensured.

ATTACHING / HAULING OF DEAD LOCOMOTIVES BY MAIL / EXPRESS / SUPERFAST / PASSENGER TRAINS

◇ 1. Number of Dead Locomotives Permitted

i) Only ONE dead locomotive can be attached

- Whether diesel or electric, only **one** dead loco is allowed in a Mail, Express, Superfast, or Passenger train.
- This is because:
 - Additional dead locomotives increase **load, drag, and braking distance**.
 - Handling multiple dead locos in a high-speed passenger train is unsafe.
 - Emergency braking performance may be affected.
- Not permitted by Rajadhani & Shatabdhi and other special trains.

◇ 2. Train Brake Power Requirement

ii) The brake power of the train must be 100% *excluding* the dead locomotive

- The dead locomotive **does not contribute** to train braking if its brakes are cut out.
- Therefore, the **remaining coaching stock must provide full 100% brake power**.
- This ensures:
 - Safe braking distance
 - Compliance with brake power rules for passenger trains
 - No compromise in passenger safety

In simple terms:

Even if the dead loco has no brakes, the *rest of the train* must have full brake power.

ATTACHING / HAULING OF DEAD LOCOMOTIVES BY GOODS TRAINS

Goods trains handle freight at moderate speeds and often operate with multiple locomotives. Because of this, the rules for attaching dead locomotives differ from passenger trains.

◇ 1. Maximum Number of Locomotives Permitted

✓ Maximum of 3 locomotives permitted

- 2 working locomotives + 1 dead locomotive
- These 3 can be marshalled together with the train load.

This is permitted only if:

✓ Brakes of the dead locomotive are operational

- The dead loco must have working air brakes
- Proportionate brakes must function
- BP/FP continuity must be maintained
- If brakes are inoperative, this rule does NOT apply.

◇ 2. Section Must Permit Double / Triple Heading

The section where the movement occurs must allow:

- Double heading, and
- Triple heading (in case two working + one dead loco)

This depends on:

- Gradient
- Bridge load restrictions
- Sectional speed
- Locomotive axle load

If triple-heading is not allowed, the dead loco cannot be hauled with the load.

 ESCORTING OF DEAD LOCOMOTIVES

Escorting refers to deploying a crew member or technical staff on the dead locomotive during movement.

◇ 1. WHEN ESCORTING IS *NOT* REQUIRED

Escorting of dead Diesel/Electric locomotives is NOT required if:

(a) Brakes of the dead locomotive are fully operational

- Brake cylinder

- Brake rigging
- Proportionate braking
- Air brake continuity all functioning

AND

(b) Dead locomotive is attached next to the train engine

Condition:

Live Loco → Dead Loco → Rest of Train

In this formation:

- Crew can visually monitor both locos
- Brake pipe pressure can be checked easily
- No need for a separate staff member

◇ 2. WHEN ESCORTING IS REQUIRED

Escorting must be provided if any of the following apply:

✓ (i) Dead loco is placed in the rear of the brake van

Formation:

Live Loco → Load → Brake Van → Dead Loco

Here, dead loco is far from the crew → escort necessary.

✓ (ii) Dead loco has defects in undergear equipment

Examples:

- Brake rigging defect
- Slack adjuster not operating
- Broken suspension components
- Defective safety brackets
- Coupler/buffer defect

Such locomotives require technical monitoring → escort mandatory.

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